

AI/ML Applications on Accelerators

A demonstration for the final presentation

Bayesian Optimization of a Free-Electron Laser

J. Duris¹,¹ D. Kennedy,^{1,2} A. Hanuka¹,¹ J. Shtalenkova¹,¹ A. Edelen,¹ P. Baxevanis,¹ A. Egger¹,¹
T. Cope,¹ M. McIntire,³ S. Ermon,³ and D. Ratner¹

¹*SLAC National Accelerator Laboratory, Menlo Park, California 94025, USA*

²*Department of Physics, University of California, Santa Cruz, Santa Cruz, California 95064, USA*

³*Department of Computer Science, Stanford University, Stanford, California 94305, USA*



(Received 13 September 2019; revised manuscript received 1 February 2020; accepted 19 February 2020; published 25 March 2020)

The Linac coherent light source x-ray free-electron laser is a complex scientific apparatus which changes configurations multiple times per day, necessitating fast tuning strategies to reduce setup time for successive experiments. To this end, we employ a Bayesian approach to maximizing x-ray laser pulse energy by controlling groups of quadrupole magnets. A Gaussian process model provides probabilistic predictions for the machine response with respect to control parameters, enabling a balance of exploration and exploitation in the search for the global optimum. We show that the model parameters can be learned from archived scans, and correlations between devices can be extracted from the beam transport. The result is a sample-efficient optimization routine, combining both historical data and knowledge of accelerator physics to significantly outperform existing optimizers.

Things to consider when reading an academic research paper

- What is the research problem?
 - Motivation
 - Background (literature review)
- What is the proposed approach?
 - Can you reproduce the results given the same resources?
- What evidence supports the conclusion?
 - Results (figures, tables)
 - Conclusion statement
- What is to follow up?
 - Remaining issues?
 - Future work

- What is the research problem?
 - Motivation
 - Background (literature review)

A simplex method for function minimization

By J. A. Nelder and R. Mead†

A simplex in n -dimensional space is a geometric shape with $n+1$ vertices.

A method is described for the minimization of a function of n variables, which depends on the comparison of function values at the $(n + 1)$ vertices of a general simplex, followed by the replacement of the vertex with the highest value by another point. The simplex adapts itself to the local landscape, and contracts on to the final minimum. The method is shown to be effective and computationally compact. A procedure is given for the estimation of the Hessian matrix in the neighbourhood of the minimum, needed in statistical estimation problems.

- What are some disadvantages of existing optimization methods?
 - Converges slowly for high-dimensional problems
 - Does not exploit prior knowledge or archived data
 - Can get stuck in local minima
- How is the specific research problem at hand affected by these disadvantages?

- What is the proposed approach?

Bayesian Optimization is a framework that combines a **probabilistic surrogate model** with an **acquisition function** to efficiently search for an optimum; Gaussian Processes are a popular choice for the surrogate model, as one implementation of BO.

A GP is a popular choice of model for Bayesian optimization [18]. Whereas a Gaussian distribution is characterized by a mean and covariance $y \sim N(\mu, \Sigma)$, a Gaussian process is determined by mean and covariance functions: $f(\mathbf{x}) \sim \text{GP}[\mu(\mathbf{x}), k(\mathbf{x}, \mathbf{x})]$, where $\mathbf{x}$ are possible inputs to the objective. The mean $\mu(\mathbf{x})$ is a function encoding the expected value of the objective function. Prior to updating the model with observations, $\mu(\mathbf{x})$ may encode prior understanding of the objective, such as a fit to

Model-Free Optimization

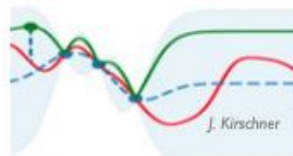


Observe performance change after a setting adjustment

→ estimate direction or apply heuristics toward improvement

gradient descent
simplex
ES

Model-guided Optimization



Update a model at each step

→ use model to help select the next point

Bayesian optimization
reinforcement learning

The Gaussian Process (GP)

A GP defines a distribution of functions

$$f(x) \sim \text{GP}[\mu(x), k(x, x')]$$

The mean function:

$$\mu(x) = \mathbb{E}[f(x)]$$

The covariance (kernel):

$$k(x, x') = \mathbb{E}[(f(x) - \mu(x))(f(x') - \mu(x'))]$$

The Gaussian Process (GP)

Given a set of measurements:

$$X = [x^{(1)}, x^{(2)}, \dots, x^{(N)}]$$

Calculate the kernel

$$K = k(X, X) + \sigma_{\text{noise}}^2 I$$

$$K = \begin{bmatrix} k(x^{(1)}, x^{(1)}) + \sigma_{\text{noise}}^2 & k(x^{(1)}, x^{(2)}) & \dots & k(x^{(1)}, x^{(N)}) \\ k(x^{(2)}, x^{(1)}) & k(x^{(2)}, x^{(2)}) + \sigma_{\text{noise}}^2 & \dots & k(x^{(2)}, x^{(N)}) \\ \vdots & \vdots & \ddots & \vdots \\ k(x^{(N)}, x^{(1)}) & k(x^{(N)}, x^{(2)}) & \dots & k(x^{(N)}, x^{(N)}) + \sigma_{\text{noise}}^2 \end{bmatrix}$$

Add in the new point

$$k(x, X) = \begin{bmatrix} k(x, x^{(1)}) \\ k(x, x^{(2)}) \\ \vdots \\ k(x, x^{(N)}) \end{bmatrix}$$

The Gaussian Process (GP)

Calculate the new mean

$$\mu(x) = k(x, X)^T K^{-1} y$$

“weight”

Calculate the new uncertainty

$$\sigma^2(x) = k(x, x) - k(x, X)^T K^{-1} k(X, x)$$

uncertainty without data

Uncertainty removed by having data

Calculate upper confidence boundary:

$$UCB(x) = \mu(x) + \kappa \sigma(x)$$

high $\mu \rightarrow$ exploitation
high $\sigma \rightarrow$ exploration

Choose the next point:

$$x_{t+1} = \arg \max_x UCB_t(x)$$

The Gaussian Process (GP)

Radial basis function (RBF) kernel:

$$k_{\text{RBF}}(x, x') = \sigma_f^2 \exp\left(-\frac{1}{2}(x - x')^T \Sigma (x - x')\right)$$

If x is just a scalar for each measurement

$$k(x, x') = \sigma_f^2 \exp\left(-\frac{(x - x')^2}{2\ell^2}\right)$$

If $|x - x'|$ is **much smaller** than $\ell \rightarrow$ kernel close to 1 (very correlated)

If $|x - x'|$ is **much larger** than $\ell \rightarrow$ kernel ≈ 0 (uncorrelated)

If x is a $m \times 1$ vector for each measurement

$$\Sigma = \text{diag}(1/\ell_1^2, \dots, 1/\ell_m^2)$$

$$k(x, x') \propto \exp\left(-\frac{1}{2} \sum_{i=1}^m \frac{(x_i - x'_i)^2}{\ell_i^2} \right) \quad \text{“length scale”}$$

Q: what is the dimension of x , X , Σ , k , K ?

The Gaussian Process (GP)

Q: what if the new measurement matches exactly one of the past measurements?

$$x_* = x_j$$

$$k_* = \begin{bmatrix} k(x_*, x_1) \\ k(x_*, x_2) \\ \vdots \\ k(x_*, x_n) \end{bmatrix} \longrightarrow k_* = \begin{bmatrix} k(x_j, x_1) \\ k(x_j, x_2) \\ \vdots \\ k(x_j, x_n) \end{bmatrix}$$

K doesn't change because K only depends on past measurements, but the j-th column of K is exactly k^* (ignoring noise first).

$$k_* = K e_j$$

← A basis vector with all 0's except 1 at position j.

The Gaussian Process (GP)

Calculate the new mean:

$$\mu(x_*) = k_*^T K^{-1} y$$

$$\mu(x_*) = (K e_j)^T K^{-1} y = e_j^T K^T K^{-1} y \xrightarrow{\text{K symmetric}} \mu(x_*) = e_j^T y = y_j$$

Calculate the new sigma:

$$\sigma^2(x_*) = \underbrace{k(x_*, x_*)}_{\sigma_f^2} - \underbrace{k_*^T K^{-1} k_*}_{\text{boxed}} = 0$$

$$\begin{aligned} k_*^T K^{-1} k_* &= (K e_j)^T K^{-1} (K e_j) = e_j^T K^T K^{-1} K e_j \\ &= e_j^T K e_j = K_{jj} = \sigma_f^2 \end{aligned}$$

Q1: how does adding measurement noise change things?

Q2: what happens when the new data is very very far from previous measurements?

The specific research problem: using GP to tune the 12 quad strengths to optimize FEL pulse energy.

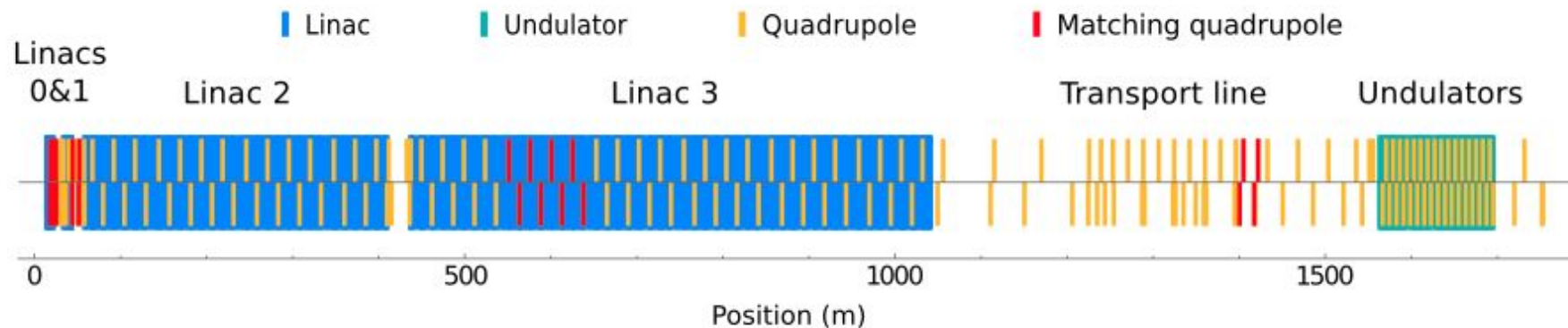


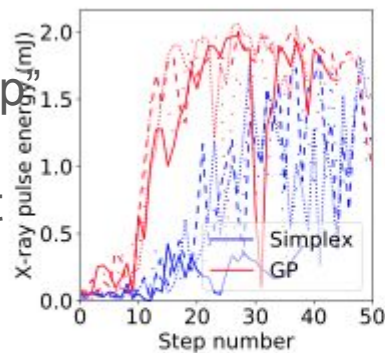
FIG. 1. LCLS beam line showing locations of linacs, quadrupole magnets, and undulator magnets. A focusing system comprised of nearly 200 quadrupoles (upper lines denote focusing whereas lower lines denote defocusing for one plane) transports the electron beam downstream (from left to right above) and several groups of matching quadrupole magnets (shown in red) are routinely used to match the beam into the focusing system. An electron beam dump and x-ray pulse energy measurement system (not shown) are positioned downstream of the undulators. Our studies focus on eight matching quads in the third linac and four in the transport line leading to the undulators.

What evidence supports the conclusion?

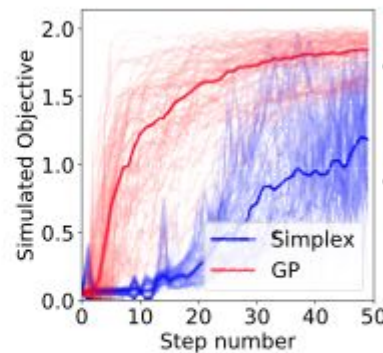
Results (figures, tables)

Conclusion statement

- What does each “step” involve?
- What do the different curves mean?
- What can you conclude from this figure?



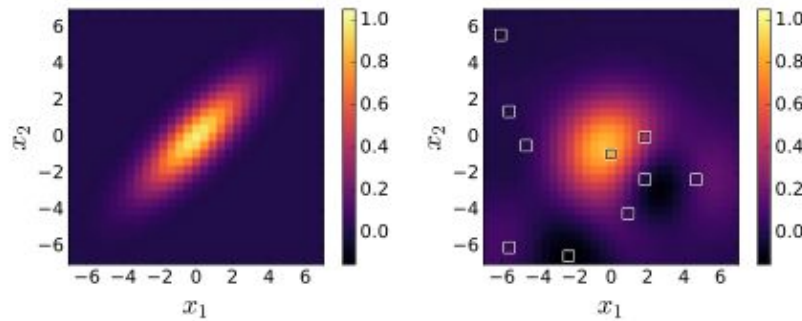
(a) Live 12 quadrupole optimization



(b) Simulated 12 quadrupole optimization

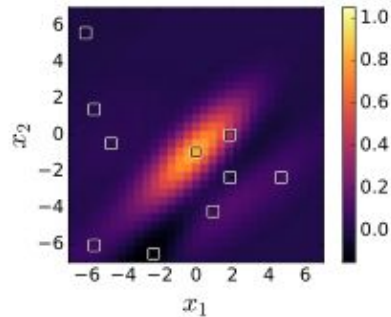
- Does simulation agree with live scans?
- How did they conduct the simulations?

We also compare simplex and Bayesian optimization in a simulation environment. Ideally we would use physics codes such as elegant [30] and Genesis [31] to model the transport and FEL behavior, however due to mismatch between the codes and measured performance, as well as computational expense for each simulation, we instead fit a beam transport model as described below. Though the beam transport model does not capture the full complexity of the real machine, it allows us to compare the relative performance of simplex and Bayesian optimization with a simulated objective function, which we find consistent with live scans [Fig. 2(b)].

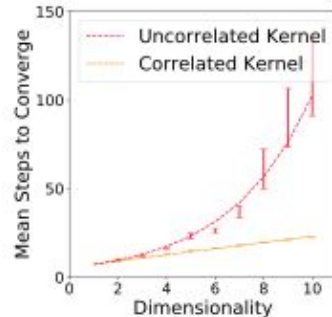


(a) Ground truth

(b) Isotropic kernel



(c) Correlated kernel



(d) Convergence tests

Simulation model: 2 quads x_1, x_2 , FEL pulse energy as the objective.

x_1 and x_2 are highly correlated

Fig3 (b) uses an **uncorrelated** kernel

Fig3 (c) uses a **correlated** kernel

$$k_{\text{RBF}}(x, x') = \sigma_f^2 \exp\left(-\frac{1}{2}(x - x')^T \Sigma (x - x')\right)$$

uncorrelated: $\Sigma = \text{diag}(1/\ell_1^2, \dots, 1/\ell_m^2)$

correlated: $\Sigma = \ell^2 \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}$

Fig3 (d) increase dimensionality from 2 to 10.

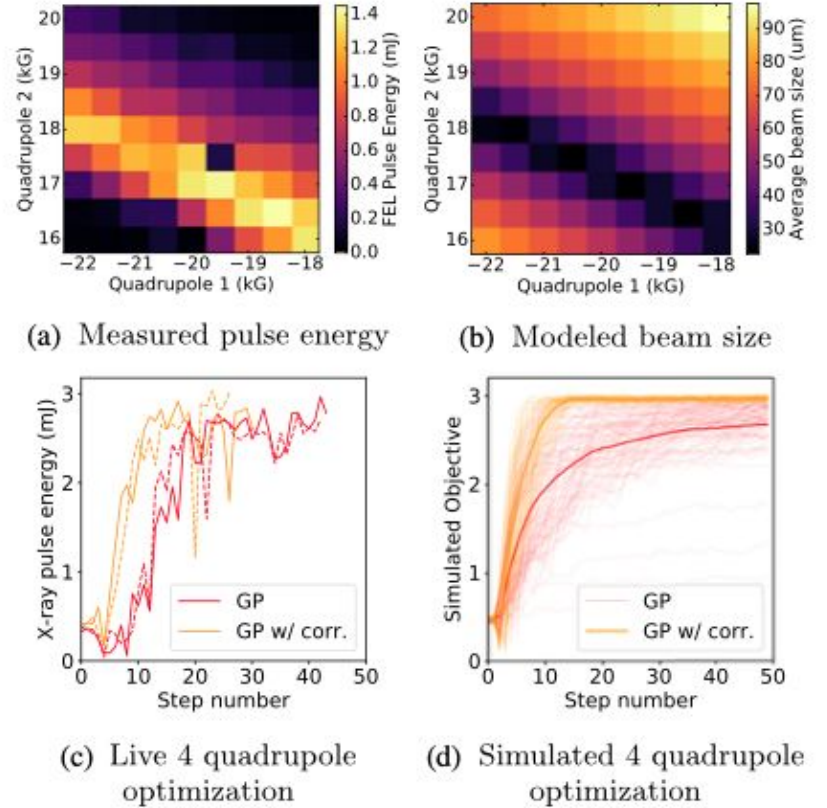
Implication: the correlated kernel performs better than the uncorrelated (incorrect) kernel. Does the FEL perform with correlation or without?

FEL quadrupole optimization is an example of a highly correlated system. Strong focusing in charged particle transport relies on a series of oppositely polarized quadrupole fields [32]. Each quadrupole focuses in one transverse plane while defocusing in the orthogonal plane, and repeated application of alternate focusing and defocusing results in net focusing in both planes. Increasing the strength of one quadrupole field necessitates increasing the strength of the next quadrupole (with opposite sign) to achieve net focusing, resulting in negative correlations between nearby focusing elements. Figure 4(a) shows the average FEL pulse energy response to variation of

Relation between FEL pulse energy and beam size:

$$\log U \propto \langle \sigma^2 \rangle^{-1/3}$$

Fig4 (b) modeled from archived quad data.
Build the correlated and uncorrelated kernel.



FEL quadrupole optimization is an example of a highly correlated system. Strong focusing in charged particle transport relies on a series of oppositely polarized quadrupole fields [32]. Each quadrupole focuses in one transverse plane while defocusing in the orthogonal plane, and repeated application of alternate focusing and defocusing results in net focusing in both planes. Increasing the strength of one quadrupole field necessitates increasing the strength of the next quadrupole (with opposite sign) to achieve net focusing, resulting in negative correlations between nearby focusing elements. Figure 4(a) shows the average FEL pulse energy response to variation of

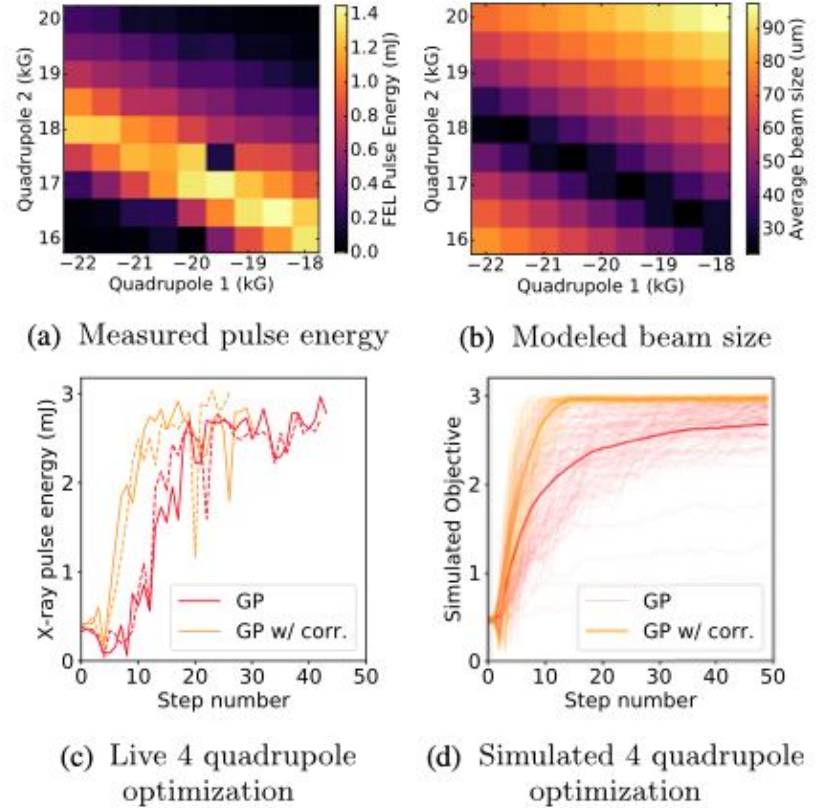
Relation between FEL pulse energy and beam size:

$$\log U \propto \langle \sigma^2 \rangle^{-1/3}$$

Fig4 (b) modeled from archived quad data.
Build the correlated and **uncorrelated** kernel.

$$\Sigma = \text{diag}(1/\ell_1^2, \dots, 1/\ell_m^2)$$

They get the length scales from fitting the archived data.



FEL quadrupole optimization is an example of a highly correlated system. Strong focusing in charged particle transport relies on a series of oppositely polarized quadrupole fields [32]. Each quadrupole focuses in one transverse plane while defocusing in the orthogonal plane, and repeated application of alternate focusing and defocusing results in net focusing in both planes. Increasing the strength of one quadrupole field necessitates increasing the strength of the next quadrupole (with opposite sign) to achieve net focusing, resulting in negative correlations between nearby focusing elements. Figure 4(a) shows the average FEL pulse energy response to variation of

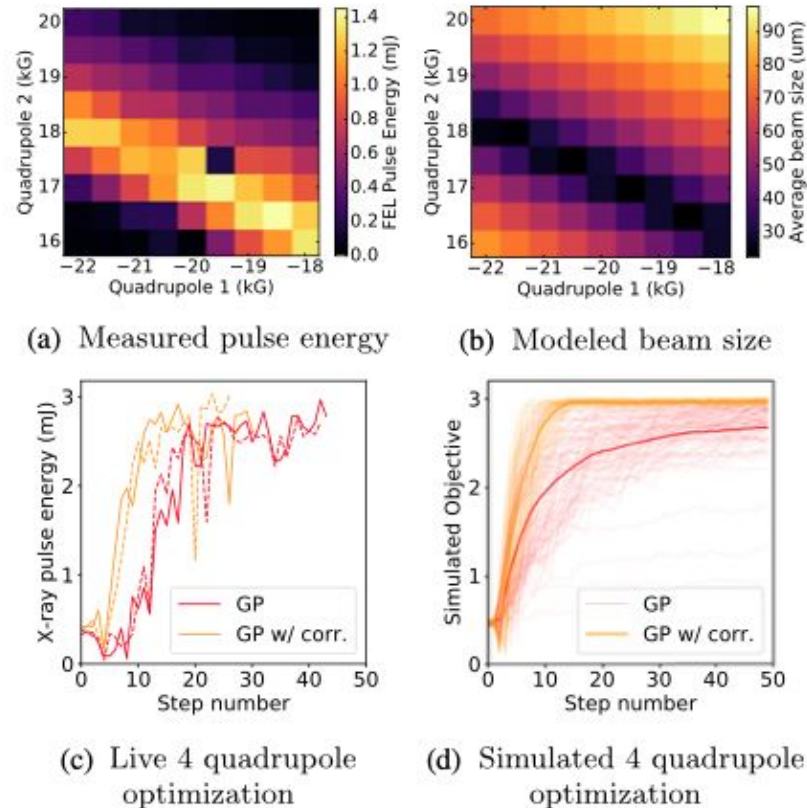
Relation between FEL pulse energy and beam size:

$$\log U \propto \langle \sigma^2 \rangle^{-1/3}$$

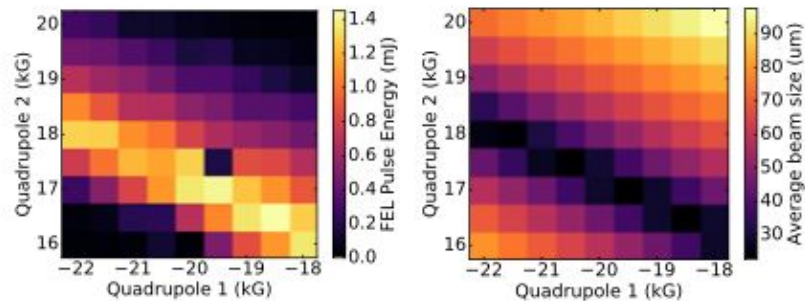
Fig4 (b) modeled from archived quad data.
Build the **correlated** and uncorrelated kernel.

$$\Sigma = \begin{bmatrix} \ell_1 & 0 & \cdots \\ 0 & \ell_2 & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix} R \begin{bmatrix} \ell_1 & 0 & \cdots \\ 0 & \ell_2 & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

They get the correlations R from the linear model used in Fig4(b).

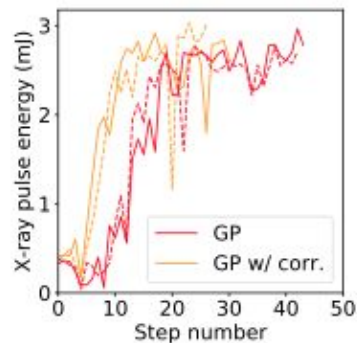


FEL quadrupole optimization is an example of a highly correlated system. Strong focusing in charged particle transport relies on a series of oppositely polarized quadrupole fields [32]. Each quadrupole focuses in one transverse plane while defocusing in the orthogonal plane, and repeated application of alternate focusing and defocusing results in net focusing in both planes. Increasing the strength of one quadrupole field necessitates increasing the strength of the next quadrupole (with opposite sign) to achieve net focusing, resulting in negative correlations between nearby focusing elements. Figure 4(a) shows the average FEL pulse energy response to variation of

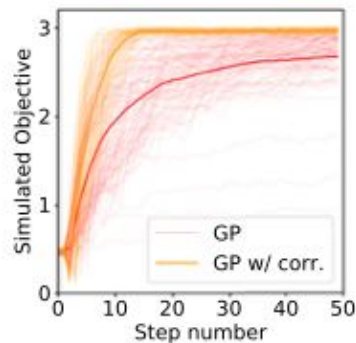


(a) Measured pulse energy

(b) Modeled beam size



(c) Live 4 quadrupole optimization

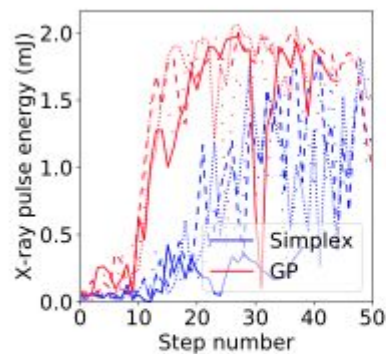


(d) Simulated 4 quadrupole optimization

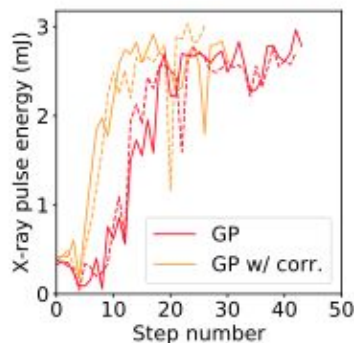
Fig4 (c,d) 4 adjacent quads

Fig4 (d) is with simulations based on correlations from the transport model.

beam transport model as shown in Fig. 4(d). It is interesting to note that the Bayesian optimizer maximized the FEL with a similar number of steps for the 12 quadrupole case [Fig. 2(a)], seemingly in contradiction to Fig. 3(d). The similarity is due to the fact that to leading order (assuming a monoenergetic beam and linear optics), only four quadrupole magnets are needed to match the four Twiss parameters into the undulator line. However, optimizing more



(a) Live 12 quadrupole optimization

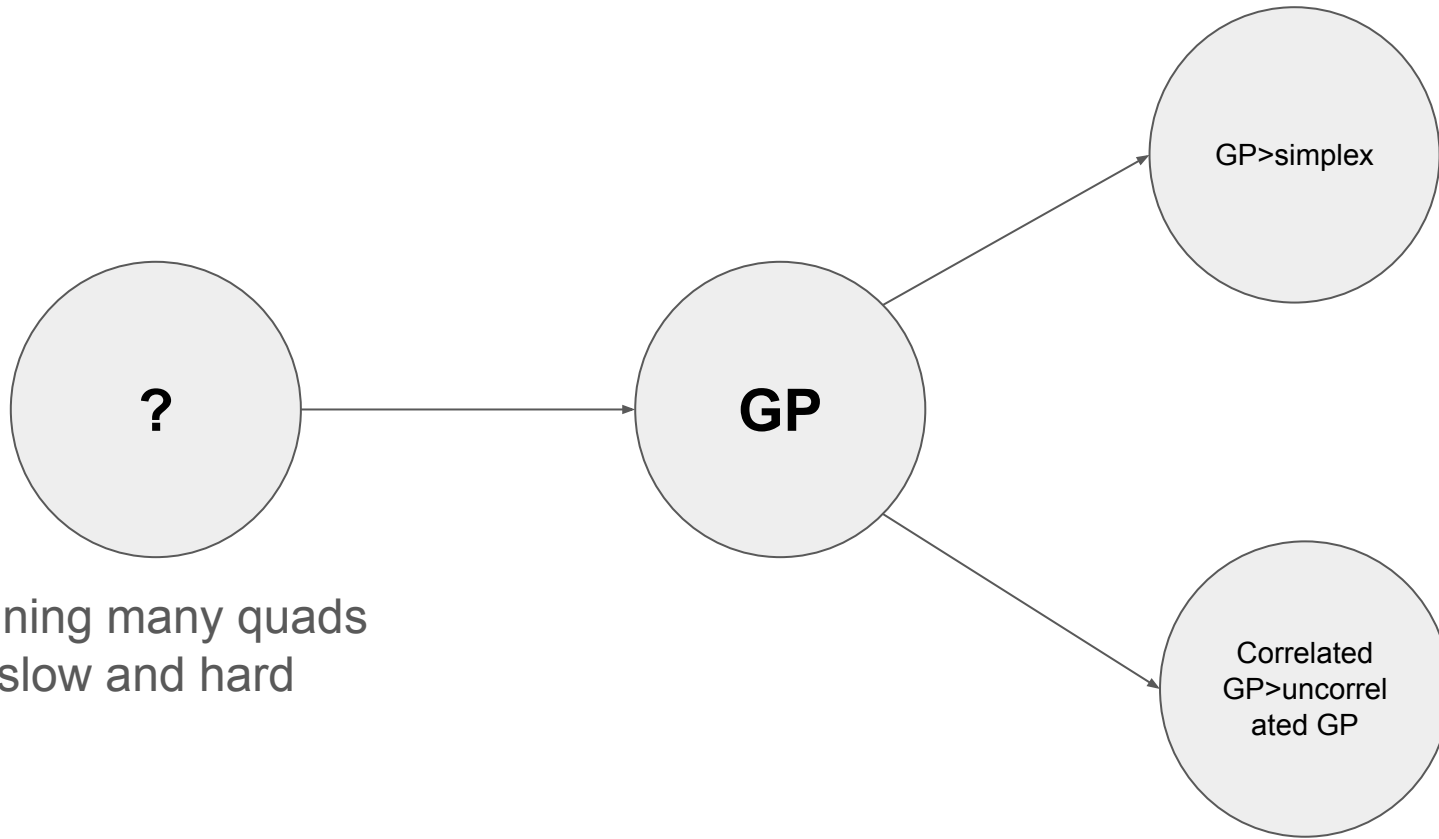


(c) Live 4 quadrupole optimization

- What is to follow up?

- Remaining issues?
- Future work

1. Due to sparsity in archived data, correlations were calculated from a simple linear transport model.
2. The simple model does not handle complications such as FEL saturation.
3. Scaling the GP to >12 quads.
4. Apply to more objective functions.
5. Include “safety” constraints.



Tuning many quads
is slow and hard

Machine learning for orders of magnitude speedup in multiobjective optimization of particle accelerator systems

Auralee Edelen^{1,*}, Nicole Neveu¹, Matthias Frey², Yannick Huber²,
Christopher Mayes¹, and Andreas Adelmann^{2,†}

¹*SLAC National Laboratory, Menlo Park, 94025 California, USA*

²*Paul Scherrer Institut, 5232 Villigen, Switzerland*



(Received 2 October 2019; accepted 24 January 2020; published 8 April 2020)

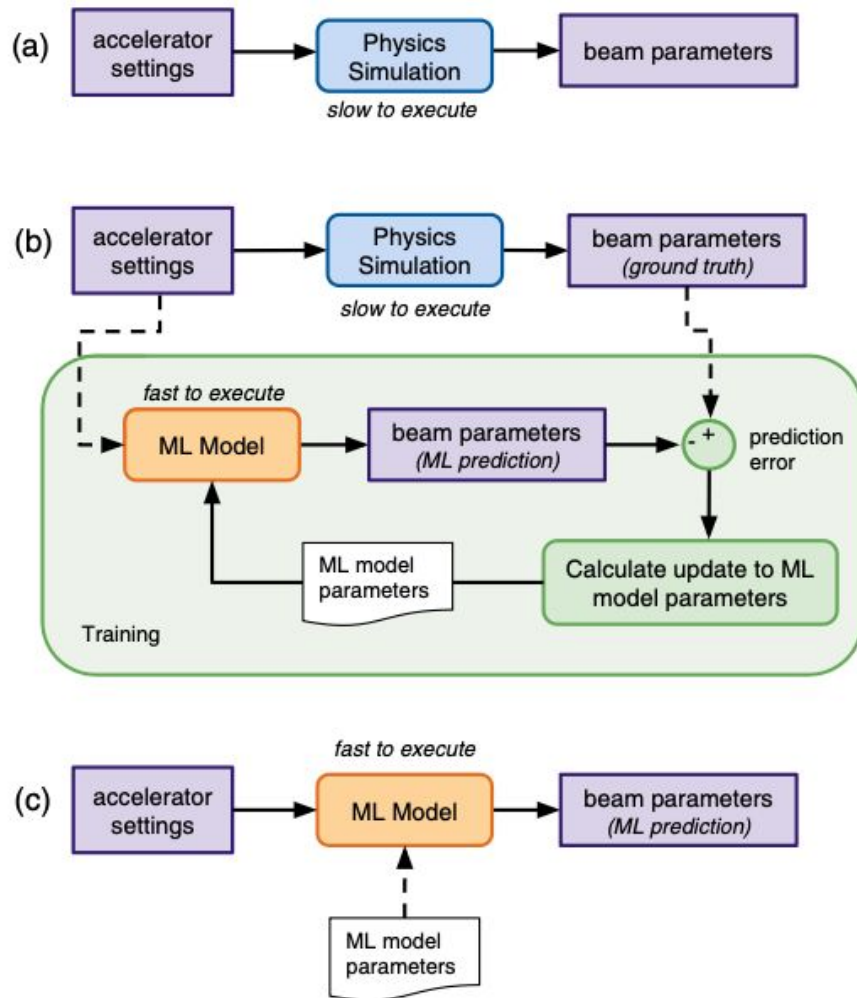
High-fidelity physics simulations are powerful tools in the design and optimization of charged particle accelerators. However, the computational burden of these simulations often limits their use in practice for design optimization and experiment planning. It also precludes their use as on-line models tied directly to accelerator operation. We introduce an approach based on machine learning to create nonlinear, fast-executing surrogate models that are informed by a sparse sampling of the physics simulation. The models are $\mathcal{O}(10^6)$ – $\mathcal{O}(10^7)$ times more computationally efficient to execute. We also demonstrate that these models can be reliably used with multiobjective optimization to obtain orders-of-magnitude speedup in initial design studies and experiment planning. For example, we required 132 times fewer simulation evaluations to obtain an equivalent solution for our main test case, and initial studies suggest that between 330–550 times fewer simulation evaluations are needed when using an iterative retraining process. Our approach enables new ways for high-fidelity particle accelerator simulations to be used, at comparatively little computational cost.

Things to consider when reading an academic research paper

- What is the research problem?
 - Motivation
 - Background (literature review)
- What is the proposed approach?
 - Can you reproduce the results given the same resources?
- What evidence supports the conclusion?
 - Results (figures, tables)
 - Conclusion statement
- What is to follow up?
 - Remaining issues?
 - Future work

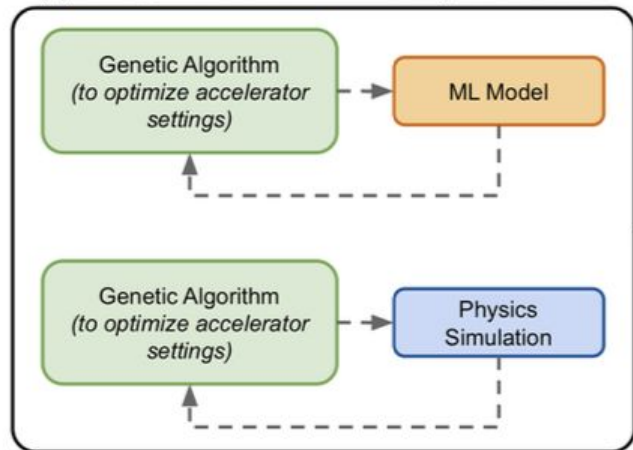
- What is the research problem?
 - High-fidelity particle-accelerator simulations (PIC, start-to-end) are very accurate but extremely slow
 - How do we keep simulation-level accuracy but run optimization thousands of times faster?
- Motivation
 - Modern accelerators (linacs, injectors, cyclotrons) involve many knobs (RF phases, gradients, solenoids, etc.) with complicated nonlinear coupling.
 - Want **multiobjective optimization** (bunch length vs emittance vs energy spread...)
 - But full simulations take thousands of CPU-hours
 - Operators cannot use simulations during live machine operation

- Background (literature review)
 - **Genetic Algorithms** widely used in accelerator tuning, but require huge simulation evaluation counts.
 - Existing speed-up attempts: parallelization, HPC
 - These help but still remain too slow for online use.
 - Prior ML uses in accelerators existed (virtual diagnostics, inverse models) but:
 - not systematically validated for multiobjective optimization
- A few words on GA
 - Works by selection -> recombination -> mutation
 - Does not require gradient (useful for nonlinear processes)
 - Can handle multiobjective optimization
 - Needs tens of thousands of iterations

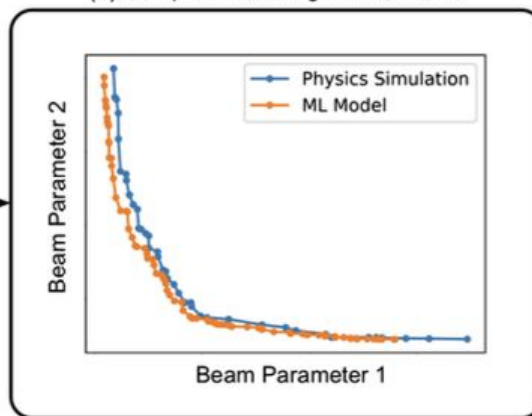


Train ML surrogate models (mainly neural networks) on a sparse random sampling of the simulation.

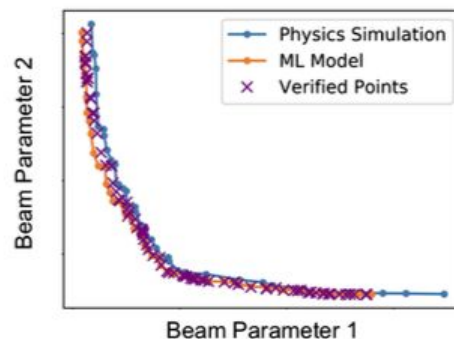
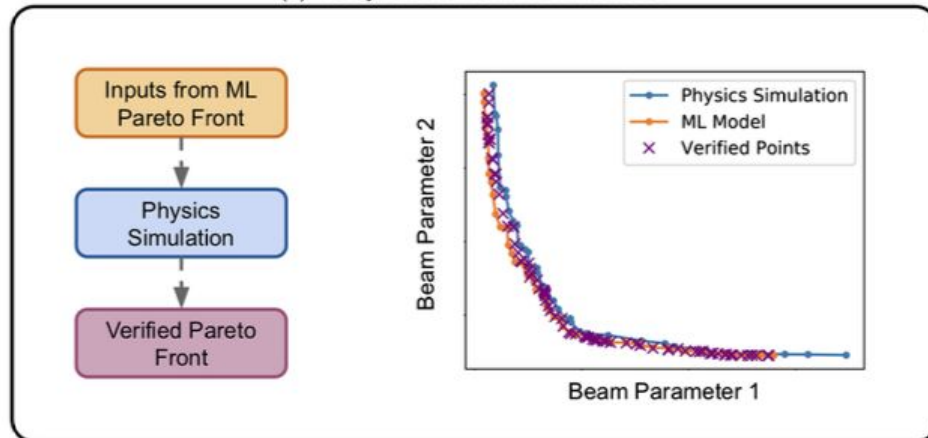
(a) Run Optimizer on ML Model and Physics Simulation



(b) Compare Resulting Pareto Fronts



(c) Verify Pareto Front from ML Model



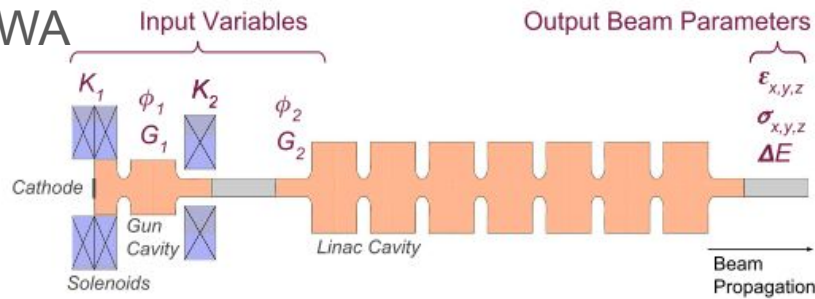
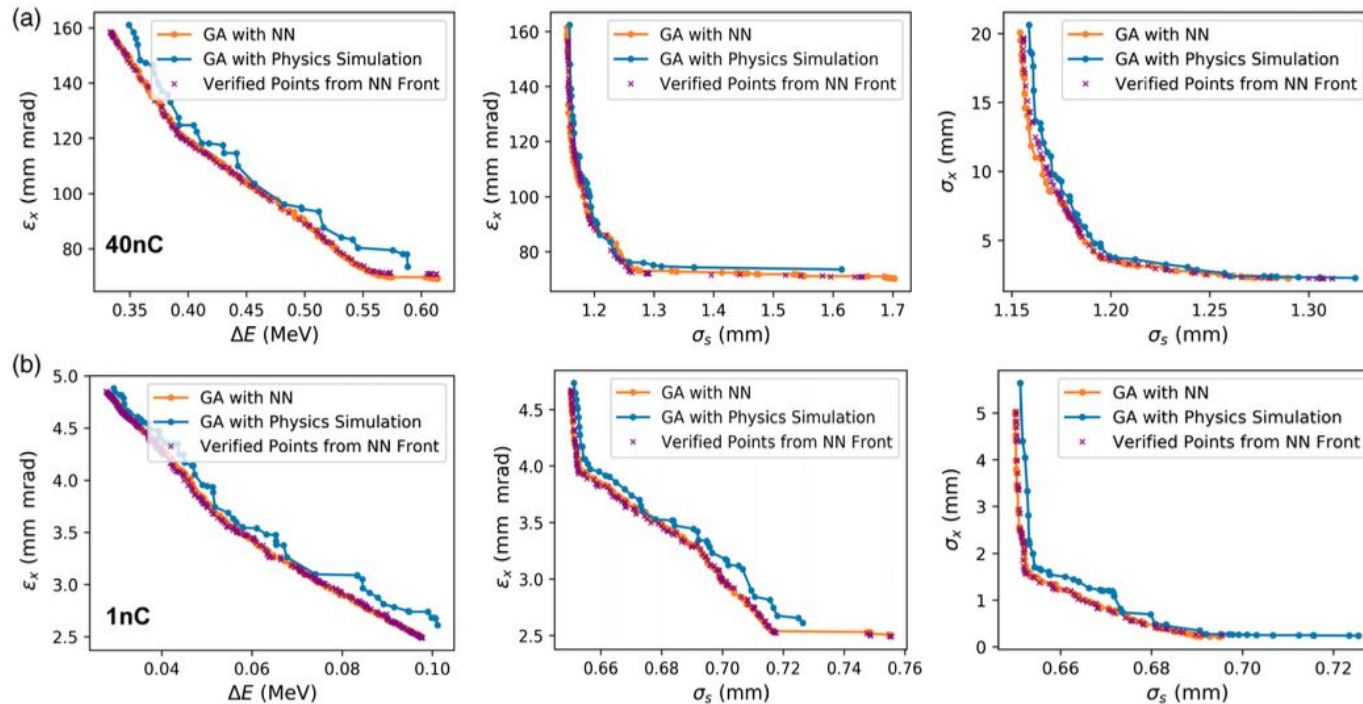


FIG. 3. Schematic of the AWA linac, together with the controllable accelerator settings and predicted beam parameters. The randomly varied inputs include the injector rf phase ϕ_1 and accelerating gradient G_1 , the linac cavity rf phase ϕ_2 and accelerating gradient G_2 , and two solenoid strengths K_1 and K_2 . The output electron beam parameters are the transverse spot sizes σ_x and σ_y , the bunch length σ_z , the transverse projected emittance values ϵ_x and ϵ_y , the longitudinal projected emittance ϵ_z , and the energy spread ΔE . The input variable ranges are determined by typical operating ranges at the AWA and are shown in Table I. We examined this setup for 40 and 1 nC bunch charges.



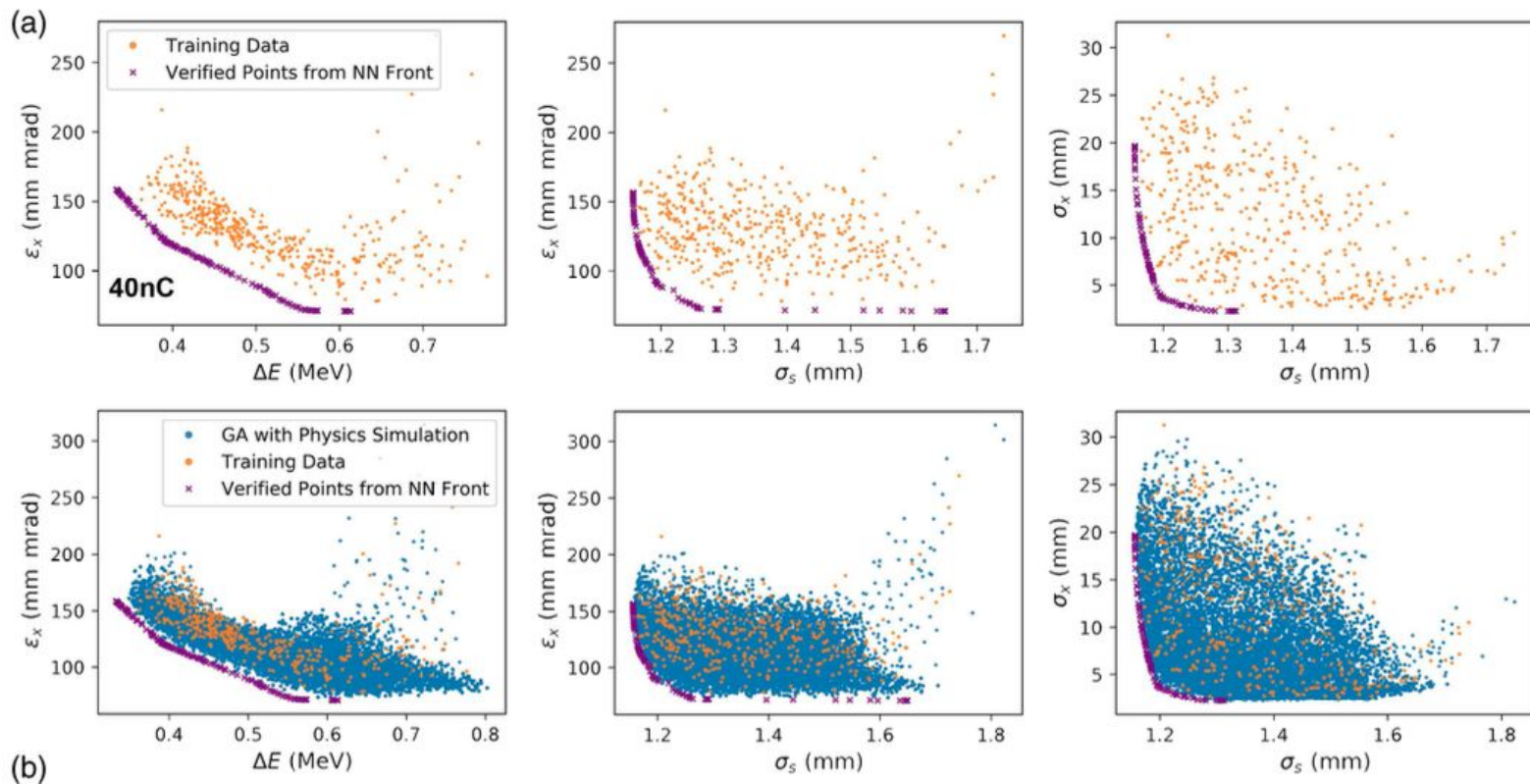
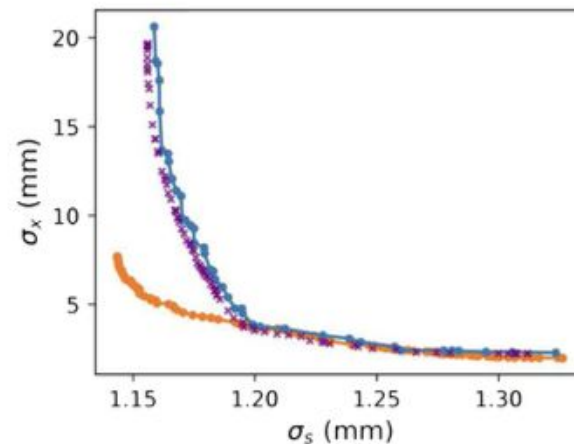
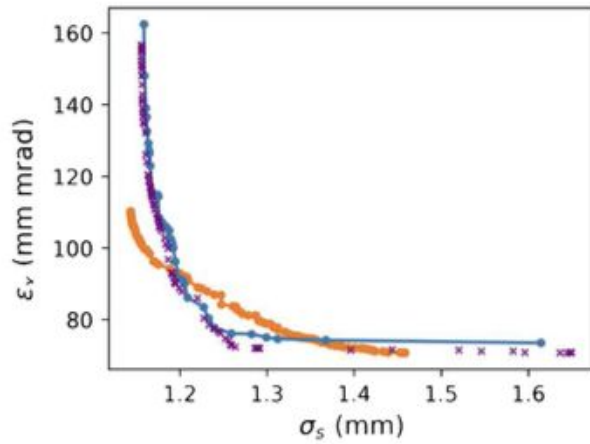
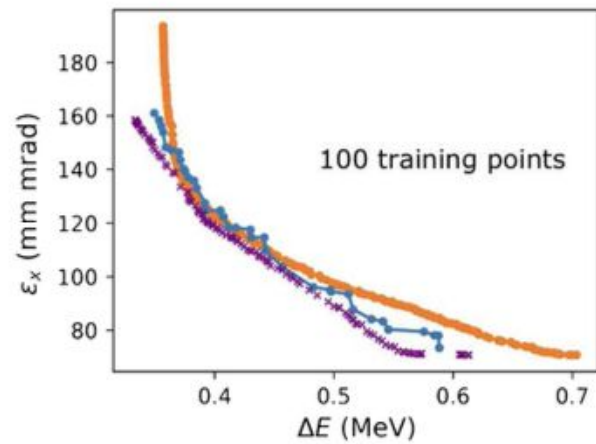
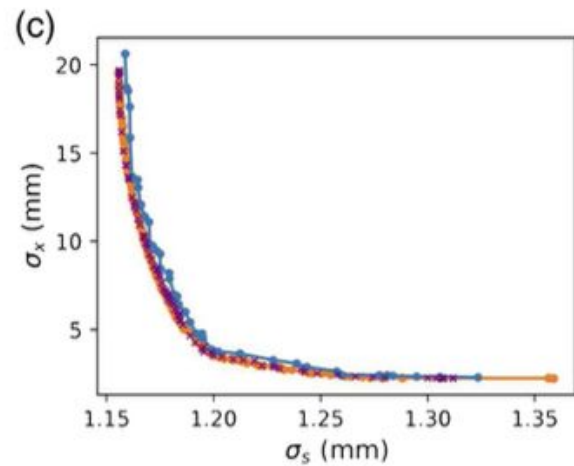
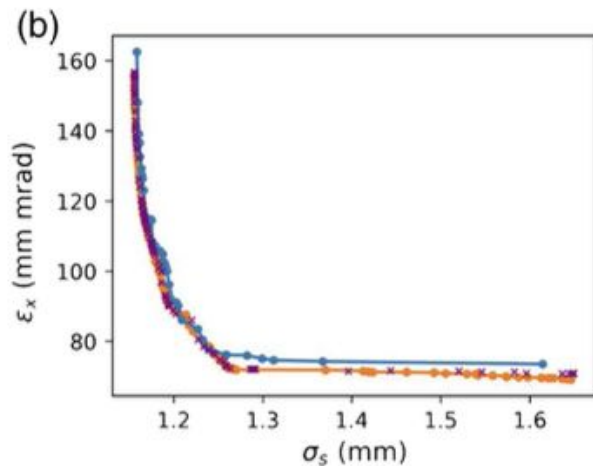
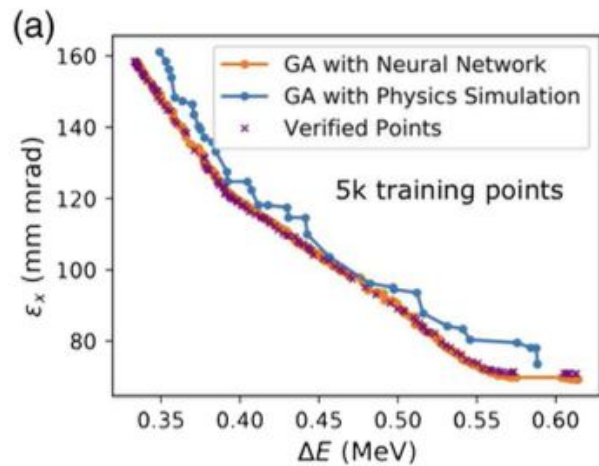
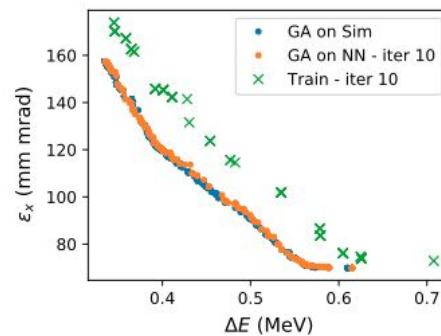
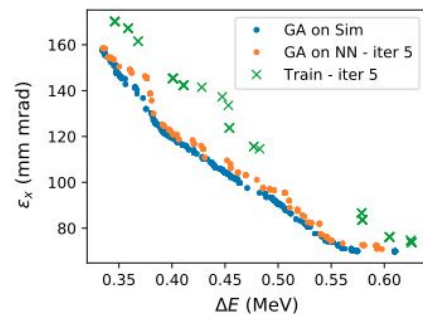
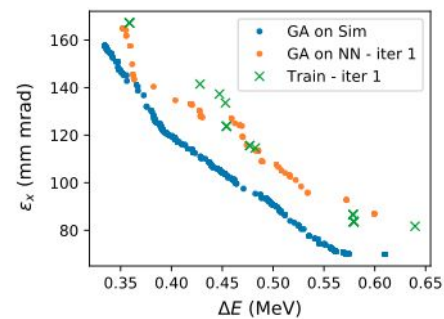
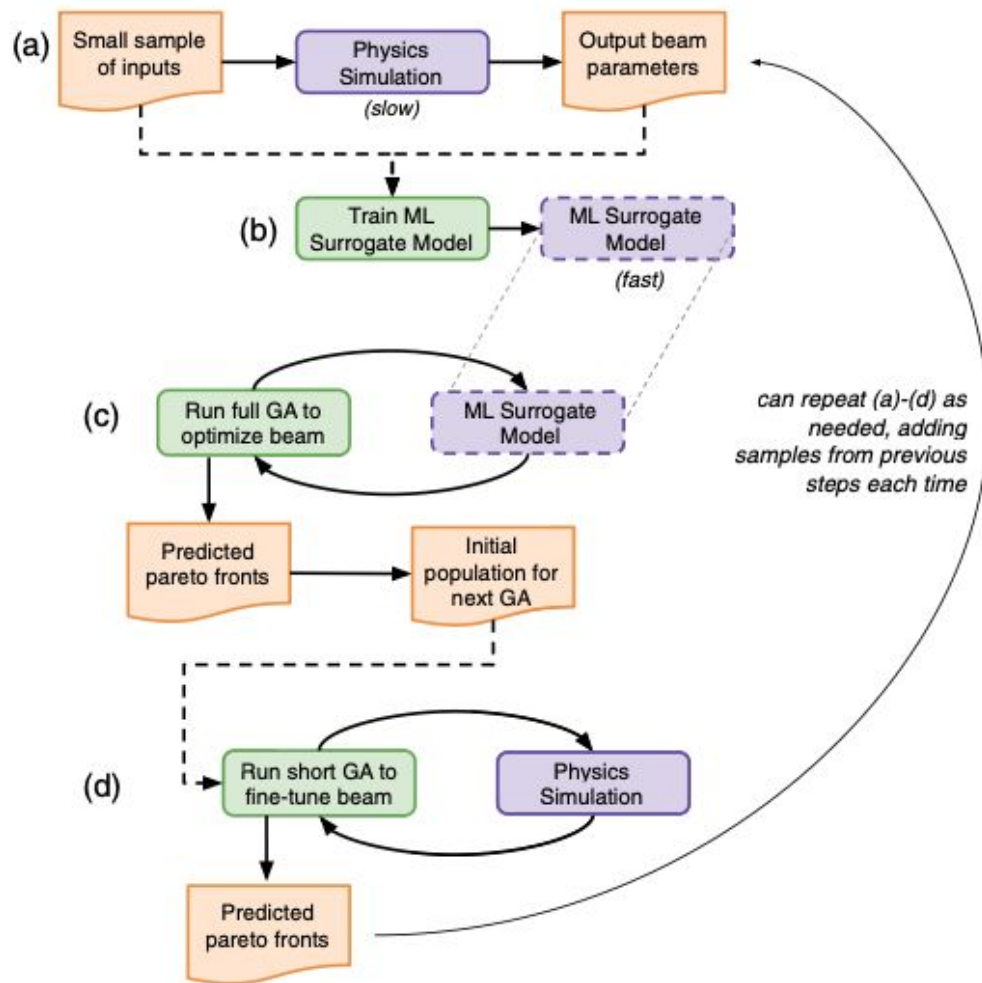


FIG. 5. Visualization of the solution found by the GA run on the NN model, as compared with both the training set and the points sampled with the GA that was run on the physics simulation. The position of the front indicates that the NN is able to interpolate in the input space to produce better combinations of output parameters than are observed in the training data. We show the sampled points in the training set and the verified points from the estimated Pareto front from the NN for 40 nC (a) and 1 nC (b).

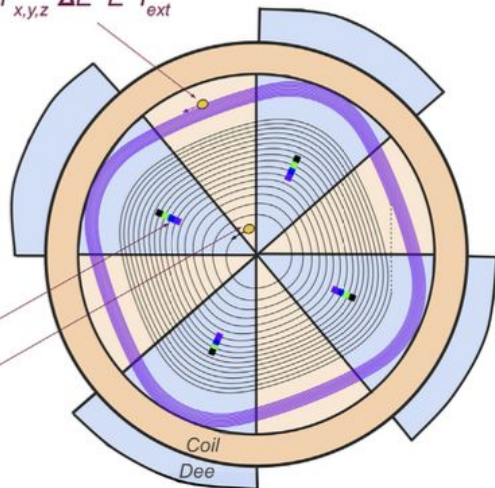




The ISODAR cyclotron

Output Beam Parameters

$\epsilon_{x,y,s}$ $\sigma_{x,y,s}$ $h_{x,y,z}$ ΔE E I_{ext}

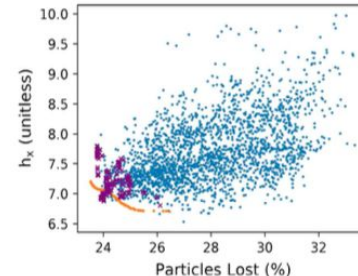
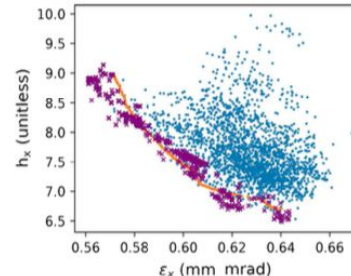
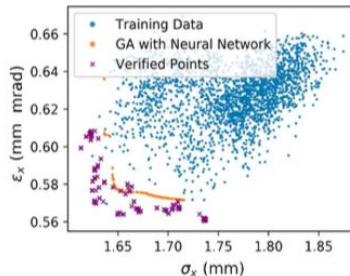


Input Variables

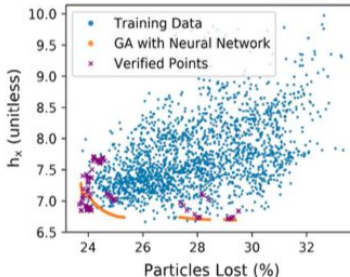
C_1 C_2 C_3 C_4

I_{inj}

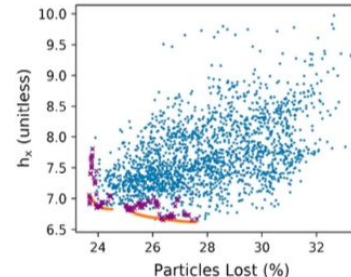
(a) 12 predicted outputs, 12 objectives used in optimization



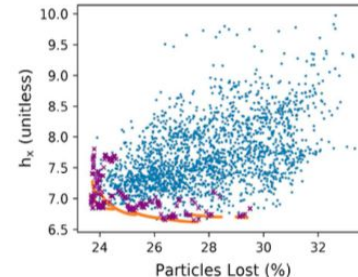
(b) 2 predicted outputs, 2 objectives



(c) Retraining with first Pareto front



(d) Cumulative result from retraining



Conclusions and outlook

- ML surrogate models can approximate accelerator-beam parameter maps
- Genetic Algorithm (GA) + ML surrogate can reproduce Pareto fronts
- Iterative retraining improves surrogate accuracy
- ML optimization reduces computational cost
- ML is promising for high-dimensional, multi-objective accelerator optimization

Remaining issues

- Struggles with highly nonlinear outputs (IsoDAR example)
- Full 12-output prediction is weak for IsoDAR

Future work

- Live updates to models, include model uncertainties, efficient sampling
- Scaling to higher dimension and complexity
- Include prior physics knowledge (GP)

Notes on the presentation

- Assume your audience have not read the paper, explain the basic concept
- Balance depth and breadth: highlight one or two major results, explain them in depth
- Not all papers are perfect. If you were a reviewer, what would you have critiqued?
- Highlight what you learned in this semester helps you understand concepts in the paper.

Rubric (each point 20%)

- What is the research problem?
- What is the proposed approach?
- What evidence supports the conclusion?
- What is to follow up?
- Highlight what you learned in this semester helps you understand concepts in the paper.